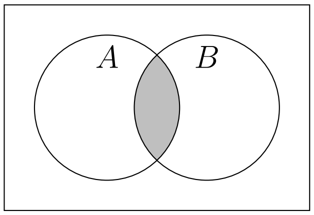
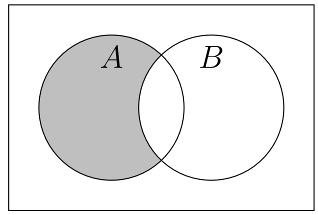
La Scuola d'Italia / Huson / IB Math: Probability First & last name:
26 February 2026

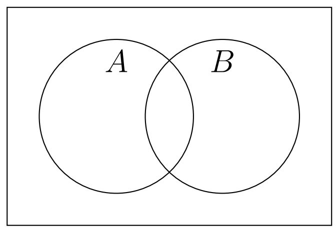
4.16 Classwork:Probability Review

1. (a) For each Venn diagram below, write an expression for the shaded region.



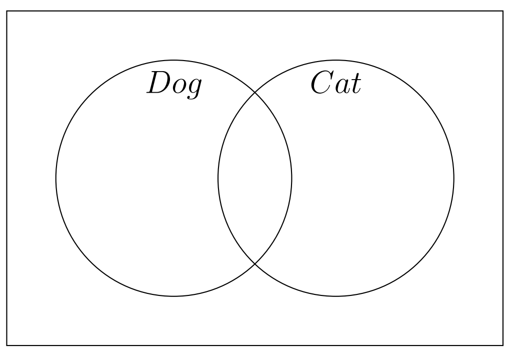


(b) Shade the region representing $A' \cup B$ in the diagram below.



2. In a survey of 50 students, 32 own a dog, 18 own a cat, and 8 own both.

(a) Complete the Venn diagram below, writing the number of students in each region.



(b) A student is selected at random. Find the probability they own a dog but not a cat.

(c) Find the probability they own neither a dog nor a cat.

3. Events A and B are such that $\mathrm{P}(A) = 0.6$, $\mathrm{P}(B) = 0.5$, and $\mathrm{P}(A \cap B) = 0.2$.

   (a) Find $\mathrm{P}(A \cup B)$. (Show substitution into a formula.)

   (b) Find $\mathrm{P}(A' \cap B')$.

   (c) Find $\mathrm{P}(A \mid B)$. (Show substitution into a formula.)

   (d) Determine whether A and B are independent. Justify your answer.

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4. Events A and B are independent, with $\mathrm{P}(A) = 0.4$ and $\mathrm{P}(B) = 0.7$.
   (a) Find $\mathrm{P}(A \cap B)$.

   (b) Complete the probability table below, including all totals.

	A	A'	Total
B			
B'			
Total			

   (c) Find $\mathrm{P}(A \cup B)$.

   (d) Find $\mathrm{P}(B \mid A')$.

5. Events A and B are mutually exclusive, with $\mathrm{P}(A) = 0.35$ and $\mathrm{P}(B) = 0.25$.
   (a) Write down $\mathrm{P}(A \cap B)$.

   (b) Find $\mathrm{P}(A \cup B)$.

   (c) Find $\mathrm{P}(A' \cap B')$.

6. In a bag of marbles, 60% are red and 40% are blue. When a red marble is selected, the probability that it is large is 0.3. When a blue marble is selected, the probability that it is large is 0.5. A marble is selected at random.

   (a) In the space below, draw a fully labelled tree diagram for this situation, including all branch probabilities.

   (b) Find the probability that the marble is red and large.

   (c) Find the probability that the marble is large.

   (d) Given that the marble is large, find the probability that it is red.